综合练习

人员

刘宸熙、纪博涵、柳力玮、蒋叔璋、刘祺 到课, 杨咏丞 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/2903

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周日三点半C++归并排序

刷新

#	用户名	姓名	编程分	时间	A	B	C	D
1	liuliwei	柳力玮	400	1104	100	100	100	100
2	zhaoshufan	赵书梵	400	1173	100	100	100	100
3	yangyongcheng	杨咏丞	400	2330	100	100	100	100
4	chujinyang	初锦阳	300	1053	100	100	100	
5	liuchenxi	刘宸熙	300	2006	100	100	100	
6	jibohan	纪博涵	300	2025	100	100	100	
7	liruihan	李瑞涵	300	2042	100	100	100	
8	gaojianhuan	高健桓	200	266	100		100	

本周作业

https://cppoj.kids123code.com/contest/3159 (课上讲了 A ~ C 题, 课后作业是 D 题)

课堂表现

今天的第三题比较难一些, 建议同学们课后好好复习一下这道题目。

课堂内容

[蓝桥杯 2015 国 AC] 穿越雷区

bfs 求最短路

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 100 + 5;
char s[maxn][maxn];
int dis[maxn][maxn];
struct node {
  int x, y;
};
int dx[] = {-1, 1, 0, 0}, dy[] = {0, 0, -1, 1};
```

```
bool check(int x, int y, char c) {
  if (s[x][y] == 'B') return true;
  if (s[x][y]=='+' && (c!='+')) return true;
  if (s[x][y]=='-' && (c!='-')) return true;
  return false;
}

int main()
{
  int n, sx, sy, ex, ey; cin >> n;
  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= n; ++j) {
      cin >> s[i][j];
      if (s[i][j] == 'A') sx = i, sy = j;
      if (s[i][j] == 'B') ex = i, ey = j;
    }
  }

  memset(dis, -1, sizeof(dis));
  queue<node> q;
  q.push({sx,sy}), dis[sx][sy] = 0;
  while (!q.empty()) {
    node u = q.front(); q.pop();
    int x = u.x, y = u.y;
    for (int i = 0; i < 4; ++i) {
      int nx = x+dx[i], ny = y+dy[i];
      if (nx>=1 && nx<=n && ny>=1 && ny<=n && check(nx,ny,s[x][y]) && dis[nx]
[ny]==-1) {
        dis[nx][ny] = dis[x][y]+1, q.push({nx,ny});
      }
    }
  }

  cout << dis[ex][ey] << endl;
  return 0;
}
```

Bitmask

位运算贪心题, 从高位往低位考虑是否要把 1 变为 0

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;

int iValue(char x) { return x-'0'; }

int main()
```

```
{
  string str; LL N; cin >> str >> N;
  reverse(str.begin(), str.end());

  LL sum = 0;
  for (int i = str.size()-1; i >= 0; --i) {
    if (str[i] != '?') sum += (1LL<<i) * iValue(str[i]);
  }

  if (sum > N) cout << -1 << endl;
  else if (sum == N) cout << N << endl;
  else {
    for (int i = str.size()-1; i >= 0; --i) {
      if (str[i] != '?') continue;
      if (sum + (1LL<<i) <= N) sum += (1LL<<i);
    }
    cout << sum << endl;
  }
  return 0;
}
```

Dist Max 2

先把所有点按照 x 进行排序, 之后二分答案, check 判断是否有两个点的距离 >= mid

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
const int inf = 0x3f3f3f3f;
struct node {
    int x, y;
    bool operator < (const node& p) const { return x < p.x; }
} w[maxn];
int n;
int suf_min[maxn], suf_max[maxn];
int x[maxn], y[maxn];

bool check(int mid) {
    for (int i = 1; i <= n; i++) {
        int pos = lower_bound(x+1, x+n+1, x[i]+mid) - x;
        if (pos == n+1) return false;
        int maxx = suf_max[pos], minn = suf_min[pos];
        if (maxx-y[i]>=mid || y[i]-minn>=mid) return true;
    }
    return false;
}

int main()
{
```

```
    cin >> n;
    for (int i = 1; i <= n; ++i) cin >> w[i].x >> w[i].y;
    sort(w+1, w+n+1);
    for (int i = 1; i <= n; ++i) x[i] = w[i].x, y[i] = w[i].y;

    suf_min[n+1] = inf, suf_max[n+1] = -inf;
    for (int i = n; i >= 1; --i) {
        suf_min[i] = min(suf_min[i+1], w[i].y);
        suf_max[i] = max(suf_max[i+1], w[i].y);
    }

    int l = 0, r = 1e9;
    while (l <= r) {
        int mid = (l + r) / 2;
        if (check(mid)) l = mid+1;
        else r = mid-1;
    }
    cout << r << endl;
    return 0;
}
```